

CORONATION, AB, Canada

WMO: 713180

Lat: 52.0742N

Lon: 111.4494W

Elev: 791

StdP: 92.18

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-29.5	-37.3	0.1	-32.3	-33.6	0.2	-29.3	13.1	-7.2	11.8	-8.1	3.2	270	0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	29.8	16.7	27.6	16.0	25.5	15.3	18.2	26.1	17.2	24.8	16.3	23.4	4.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	12.3	20.1	14.6	11.4	19.2	13.6	10.7	18.4	54.5	26.2	51.4	24.9	48.5	23.5	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	10.0	8.8	-36.7	33.1	3.4	2.1	-39.1	34.6	-41.1	35.8	-42.9	37.0	-45.4	38.5		
			-36.7	20.0	3.2	1.2	-39.0	20.9	-40.9	21.6	-42.7	22.3	-45.0	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-13.1	-11.6	-5.6	3.2	9.6	13.8	17.0	16.0	10.7	3.4	-5.3	-11.7	(6)
(7)		DBStd	12.42	8.85	7.71	7.92	5.41	4.34	3.26	3.18	3.50	4.49	5.02	6.99	8.12	(7)
(8)		HDD10.0	3472	717	605	484	210	60	6	1	2	44	211	459	674	(8)
(9)		HDD18.3	5907	975	838	742	455	270	138	64	89	230	464	708	932	(9)
(10)		CDD10.0	648	0	0	0	5	49	121	216	187	66	5	0	0	(10)
(11)		CDD18.3	42	0	0	0	0	1	3	22	16	1	0	0	0	(11)
(12)		CDH23.3	992	0	0	0	1	43	94	396	353	102	2	0	0	(12)
(13)		CDH26.7	280	0	0	0	0	6	16	126	107	26	0	0	0	(13)
(14)	Wind	WSAvg	4.2	4.3	4.2	4.4	4.7	4.6	4.2	3.8	3.7	4.0	4.3	4.2	4.1	(14)
(15)	Precipitation	PrecAvg	405	20	17	21	23	41	63	74	51	40	16	16	22	(15)
(16)		PrecMax	570	52	37	55	77	118	177	212	150	139	51	37	42	(16)
(17)		PrecMin	239	4	2	3	0	5	14	16	6	3	2	2	5	(17)
(18)		PrecStd	75	13	9	13	15	24	35	43	35	30	10	9	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	4.7	13.0	22.2	27.1	28.6	32.9	32.4	30.0	22.2	12.6	6.8	(19)
(20)		MCWB	2.0	1.7	6.4	10.9	14.1	16.4	17.7	16.7	14.9	11.4	6.4	2.7		(20)
(21)		2%	DB	3.2	2.5	9.2	18.8	24.3	26.0	30.0	29.5	26.2	17.7	8.5	4.2	(21)
(22)		MCWB	0.9	0.3	4.1	9.1	12.5	15.3	17.7	16.3	13.9	9.5	4.1	1.3		(22)
(23)		5%	DB	1.3	0.7	6.1	15.5	21.7	23.8	27.6	27.3	23.4	14.7	5.7	1.8	(23)
(24)		MCWB	-0.5	-1.0	2.4	7.3	11.7	14.4	16.9	15.9	13.1	7.9	2.2	-0.4		(24)
(25)		10%	DB	-1.0	-1.1	3.5	12.7	19.2	21.8	25.2	24.9	20.3	11.7	3.3	-0.5	(25)
(26)		MCWB	-2.3	-2.5	0.9	5.8	10.2	13.5	16.2	15.3	11.8	6.1	0.8	-2.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.1	7.1	11.2	15.2	17.9	20.1	18.8	15.9	12.2	6.7	3.2	(27)
(28)		MCDWB	4.4	4.2	12.1	20.9	25.4	25.4	28.4	26.5	25.7	20.6	11.7	6.3		(28)
(29)		2%	WB	1.0	0.4	4.4	9.4	13.4	16.4	18.7	17.8	14.7	10.0	4.4	1.5	(29)
(30)		MCDWB	3.0	2.1	8.7	18.0	21.9	23.5	26.5	25.8	24.0	16.5	8.1	4.0		(30)
(31)		5%	WB	-0.4	-0.9	2.6	7.8	12.2	15.3	17.7	16.8	13.5	8.3	2.5	-0.3	(31)
(32)		MCDWB	1.2	0.6	5.6	14.8	20.2	21.6	25.4	24.7	22.2	13.7	5.1	1.7		(32)
(33)		10%	WB	-2.4	-2.5	1.1	6.1	10.9	14.3	16.7	15.8	12.3	6.7	1.0	-2.1	(33)
(34)		MCDWB	-1.0	-1.1	3.4	11.7	18.1	20.1	23.7	23.3	19.8	11.3	3.0	-0.5		(34)
(35)	Mean Daily Temperature Range	MDBR	9.8	10.2	9.8	12.2	13.5	12.4	13.8	14.2	14.2	12.4	9.9	9.8		(35)
(36)		5% DB	MCDBR	11.5	11.5	12.7	17.8	18.0	16.3	17.9	19.0	20.1	17.6	12.9	11.3	(36)
(37)		MCWBR	9.6	9.6	8.3	9.5	8.7	7.4	7.3	7.8	9.4	10.0	8.9	8.8		(37)
(38)		5% WB	MCDBR	11.5	11.3	12.0	17.0	16.3	14.0	15.6	16.1	18.3	16.7	11.9	10.9	(38)
(39)		MCWBR	9.9	9.6	8.2	9.4	8.1	6.9	7.1	7.2	8.9	9.8	8.5	8.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.266	0.309	0.288	0.336	0.359	0.368	0.359	0.374	0.319	0.283	0.258	0.252		(40)
(41)		taud	2.300	2.208	2.449	2.343	2.324	2.341	2.406	2.334	2.459	2.540	2.471	2.325		(41)
(42)		Ebn at Noon	746	797	919	901	892	884	887	850	869	846	773	704		(42)
(43)		Edh at Noon	62	92	88	111	119	119	109	111	87	65	52	51		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.13	2.19	3.61	4.80	5.81	6.01	6.40	5.20	3.70	2.23	1.16	0.85		(44)
(45)		RadStd	0.10	0.18	0.27	0.43	0.39	0.40	0.30	0.41	0.36	0.29	0.11	0.08		(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page